

Guide 43 — Solar Photovoltaic Installer

Controlled English working master — version 2.0

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What this job is

Solar photovoltaic (PV) installers assemble, install, and maintain systems that turn sunlight into electricity. Work may include measuring and laying out an array, moving modules and racking, fastening approved components, weatherproofing penetrations within the worker's authorized scope, routing approved cables, labeling equipment, completing records, and helping test or maintain a system under the required supervision.

The title alone does not define what a worker is legally allowed to do. Employers use titles such as solar installer, PV installer, solar technician, rooftop installer, racking installer, solar crew member, electrician helper, solar electrician, and operations-and-maintenance technician. Two jobs with similar titles may have different duties, training requirements, and legal boundaries.

Mechanical module and racking work is not the same as electrical design or licensed electrical work. Grid connection, service equipment, conductor and protection sizing, grounding and bonding decisions, inspection, commissioning, energization, and utility interconnection may require a licensed electrician, qualified person, engineer, inspector, utility approval, or another authorization set by the jurisdiction and employer.

This guide is for career exploration and preparation. It does not teach a reader to install or energize a PV system independently. Verify the current local trade rules, permits, codes, employer requirements, and utility procedures before accepting responsibility for regulated work.

What you may do

Depending on the employer, project, training, and local rules, responsibilities can include:

- review approved drawings, work orders, manufacturer instructions, and site-specific plans;
- measure and mark approved locations for modules, racking, supports, or related equipment;
- stage, lift, and secure modules and components using the employer's approved handling and access methods;
- assemble and fasten racking, supports, modules, and balance-of-system components within authorization;
- assist with roof or ground-mount work after structural suitability and access controls have been confirmed by the responsible parties;
- route, support, protect, and label approved cables or components when trained and authorized;

- perform visual inspections and basic approved checks, document results, and report damage or abnormal conditions;
- use employer-approved tools, lifts, ladders, scaffolds, fall-protection systems, and material-handling equipment after training and authorization;
- keep site, equipment, inventory, photographs, serial numbers, and completion records accurate;
- communicate with supervisors, electricians, roofers, engineers, inspectors, customers, and utility representatives; and
- stop and escalate when electrical status, roof integrity, fall protection, structural suitability, weather, equipment condition, or work authorization is uncertain.

O*NET notes that PV installers may perform minor electrical work such as current checks. That occupational description is not universal permission to perform electrical work. The specific task must still be allowed by local law, the employer, the worker's qualification level, and the site procedure.

Work environment and physical demands

Installers may work on roofs, ladders, scaffolds, lifts, ground-mounted arrays, construction sites, warehouses, commercial properties, and utility-scale sites. Work can involve heat, sun, wind, cold, rain, glare, noise, sharp edges, awkward positions, lifting, carrying, kneeling, climbing, and repeated movement.

Schedules may begin early to avoid peak heat or to coordinate with other trades. Some positions require travel, a driver's license, outdoor work in changing weather, or work away from home. Maintenance roles may involve troubleshooting and documentation as well as physical installation work.

Read the actual job description and ask how often the role involves roof access, lifting, travel, electrical exposure, confined or restricted spaces, and extreme weather. Where applicable, workers may request lawful workplace accommodations, but an accommodation depends on the essential functions and safety requirements of the specific position.

Safety comes before production

PV work combines construction, work-at-height, electrical, structural, weather, and material-handling hazards. A module can generate electricity when illuminated. Turning off a building or opening one disconnect does not necessarily make every module, connector, cable, or conductor electrically safe.

Do not improvise electrical isolation, lockout/tagout, fall protection, roof access, anchors, ladders, scaffolds, lifting methods, grounding or bonding, temporary wiring, testing, commissioning, or utility interconnection. Follow the employer's approved procedures, applicable regulations and codes, the site plan, and manufacturer instructions.

Stop and escalate if:

- you cannot confirm the electrical state of the equipment or conductors;
- required qualified-person or licensed-trade supervision is absent;

- a task is outside your training, authorization, or legal scope;
- roof strength, structural suitability, access, edge protection, skylight protection, or fall-protection status is unclear;
- wind, lightning, rain, ice, heat, smoke, or another condition exceeds the employer's limits;
- a module, connector, cable, tool, ladder, scaffold, lift, harness component, or attachment point appears damaged;
- drawings, labels, equipment ratings, manufacturer instructions, or approved components do not match the work;
- a required permit, inspection, lockout, utility authorization, or site approval is missing; or
- pressure to continue conflicts with the approved safety procedure.

Falls, roof access, and structural suitability

OSHA identifies falls as a major solar-installation hazard. Workers may be exposed near roof edges, skylights, hatches, ladders, and scaffolds. In U.S. construction, 29 CFR 1926.501 generally requires fall protection for employees exposed to unprotected sides and edges 6 feet (1.8 m) or more above a lower level, subject to the standard's detailed provisions.

The same standard requires the employer to determine that a walking or working surface has the strength and structural integrity to support employees safely before work begins. A career guide cannot determine whether a particular roof, anchor, ladder, scaffold, warning-line system, guardrail, or harness arrangement is safe.

Use only the employer's approved access and fall-protection system. Do not step onto a roof, remove a guard, move an anchor, cross a warning line, or change a ladder or scaffold setup because an online diagram or AI response says it is acceptable.

Electrical and energized-PV boundaries

OSHA's solar-energy guidance identifies electrical shock, arc flash, and thermal burns among the hazards workers may face while connecting panels to electrical circuits. Which OSHA provision applies can depend on the employer and work activity. State, provincial, territorial, national, electrical-code, permit, and utility rules can add further requirements.

Do not use this guide to decide conductor size, overcurrent protection, grounding and bonding, rapid-shutdown design, service-panel changes, meter work, equipment ratings, disconnect placement, commissioning values, or interconnection settings. These decisions belong to the qualified people and authorities responsible for the project.

Treat damaged connectors, exposed conductors, mismatched equipment, water intrusion, abnormal heat, arcing signs, unexpected voltage, missing labels, and uncertain isolation as stop-work conditions. Never assume a connector is safe to open or touch merely because a switch is off.

Skills that matter

Employers commonly value:

- accurate measurement and layout;
- safe material handling and teamwork while moving modules and racking;
- ability to follow drawings, labels, checklists, and manufacturer instructions;
- basic construction math and careful use of authorized hand and power tools;
- attention to roof, weatherproofing, cable-support, equipment, and finish details;
- hazard recognition and willingness to stop when conditions are unclear;
- clear communication across installation, electrical, roofing, engineering, inspection, and customer teams;
- reliable attendance, outdoor-work readiness, and accurate documentation; and
- basic digital skills for work orders, photographs, equipment records, monitoring portals, and jobsite reporting.

Some roles also value roofing, construction, electrical-helper, warehouse, maintenance, telecom, or field-service experience. Do not describe yourself as a licensed electrician, engineer, certified installer, qualified electrical worker, inspector, or commissioning technician unless you actually hold the required credential or authorization.

Education and entry pathways

United States

The U.S. Bureau of Labor Statistics classifies Solar Photovoltaic Installers under SOC **47-2231**. BLS reports that a high school diploma or equivalent is the typical entry-level education and that workers typically receive moderate-term on-the-job training, sometimes lasting up to one year.

That national profile does not mean every employer has the same entry requirements. Some employers hire helpers and train them. Others prefer construction experience, a technical certificate, electrical coursework, fall-protection training, a driver's license, or experience with particular systems.

A practical entry sequence is:

1. Review 15–20 local postings for solar installer, PV installer, solar crew member, racking installer, electrician helper, and solar technician roles.
2. Separate mechanical installation requirements from electrical-trade requirements.
3. Compare employer-paid training, public workforce funding, technical or community college programs, and apprenticeship openings before paying a private provider.
4. Build supervised experience in measurement, material handling, racking, module placement, documentation, and jobsite safety.
5. Add electrical or specialty training only through a pathway recognized by the employer and jurisdiction for the work you want to perform.

Canada

Government of Canada Job Bank maps **solar panel installer** to **Residential and commercial installers and servicers, NOC 73200**. Its national summary says this occupational group usually requires a college diploma, an apprenticeship of less than two years, or more than six months of on-the-job training. Regional expectations differ.

Electrical work must be treated separately. The Red Seal Program identifies **Construction Electrician, NOC 72200**, as a designated Red Seal trade across participating provinces and territories. Its work can include installing, verifying, commissioning, and connecting electrical systems, and Red Seal materials identify solar systems as an emerging technology for the trade.

A solar-panel-installer title does not itself authorize work reserved to electricians or other registered or certified people. Check the provincial or territorial apprenticeship authority, compulsory-trade rules, permits, inspections, and utility requirements where you intend to work. Red Seal recognition does not override provincial or territorial law.

Colombia

SENA/Betowa lists **Mantenimiento e instalación de sistemas solares fotovoltaicos** as a **Técnico** program with a stated duration of **2,200 hours**. It also lists **Diseño y montaje de sistemas solares fotovoltaicos** as a **96-hour** complementary special course. These are different pathways and should not be treated as equivalent.

Availability is tied to current cohorts, locations, and enrollment windows. Verify the live Betowa listing before planning around a program. Neither course should be represented as a universal electrical license or automatic authorization to design, sign, inspect, energize, commission, or interconnect a system.

Electrical installations in Colombia are governed by the current **Reglamento Técnico de Instalaciones Eléctricas (RETIE)** and related requirements. The Ministry's current RETIE webpage contains inconsistent year information for Resolution 40284, so this guide does not repeat the disputed year. Use the current regulation and controlling documents published by the Ministry, then verify NTC 2050 requirements, professional competency and registration, design and signature responsibility, inspection, conformity, and distribution-network procedures for the specific project.

Latin America and the Caribbean

Training, trade, electrical, permit, and interconnection systems differ by country. OIT/Cinterfor's institutional network can help readers find national public vocational bodies such as SENA in Colombia, INA in Costa Rica, INTECAP in Guatemala, INFOTEP in the Dominican Republic, and other participating institutions.

Use the network as a locator. Confirm the current solar, electrical, construction, or renewable-energy offering with the national institution. Do not assume a course, credential, apprenticeship, or electrical authorization transfers automatically between countries.

Free-first and low-cost learning strategy

Do not assume the most expensive solar certificate is the best first step. Start with the actual local labor market:

1. Choose a target role and review real local postings.
2. Record repeated requirements for experience, construction skills, electrical credentials, safety training, travel, tools, and driving.
3. Search employer trainee roles, public technical programs, workforce funding, registered apprenticeship locators, and national vocational institutions.
4. Verify whether a program is eligible for public aid and whether local employers recognize it.
5. Ask for total tuition, equipment and exam costs, refund rules, schedule, hands-on supervision, and job-placement methodology in writing.
6. Borrow only after checking grants, public funding, employer support, paid work-based learning, scholarships, and low-cost alternatives.

Online study can help with terminology, basic electricity concepts, plans, safety awareness, and job-search preparation. It cannot replace supervised hands-on training, employer authorization, or a legally required trade credential.

Apprenticeships and work-based learning

United States

Use **Apprenticeship.gov** as a live locator. Search solar, electrical, construction, roofing, and related occupations, then verify the employer or sponsor, location, paid status, eligibility, current opening, training schedule, and recognized occupation. A search result is not a guarantee that an exact solar-installer apprenticeship is available in every state.

State employment services, technical colleges, state apprenticeship agencies, contractors, employers, and union-management apprenticeship committees may also have paid learning routes.

Canada

Check the provincial or territorial apprenticeship system for Construction Electrician and any installer-related pathway. Confirm whether the trade is compulsory, how an employer registers an apprentice, what technical training and hours apply, and whether prior experience can be assessed.

Colombia and the region

SENA and other public vocational institutions may offer technical, complementary, or employment-linked learning. Enrollment may be cohort-based. Verify admission, cost, schedule, location, credential type, work placement, and the precise scope the training supports.

Funding and employer support

Funding is conditional and must be verified before enrollment.

In the United States, an eligible college, career-school, or trade-school program may qualify for federal student aid through the FAFSA process. Not every private solar course or bootcamp is eligible. Federal loans are repayable; grants and work-study have their own eligibility rules.

American Job Centers and local workforce boards may fund eligible training through WIOA or related programs. Ask whether a specific solar, electrical, construction, or renewable-energy program is currently approved and whether you meet local eligibility rules before enrolling.

Employer support can include paid onboarding, wages during training, tuition or exam reimbursement, tools, PPE, travel, manufacturer training, apprenticeship sponsorship, or promotion into electrical, operations-and-maintenance, crew-lead, or quality roles. Get repayment, retention, wage, tool-charge, and reimbursement terms in writing.

Income and job outlook

United States — official occupational data

BLS reports for **Solar Photovoltaic Installers, SOC 47-2231**:

- **May 2024 median annual wage: \$51,860**
- **May 2024 median hourly wage: \$24.93**
- **Lowest 10%: below \$39,070**
- **Highest 10%: above \$80,150**
- **Employment in 2024: 28,600**
- **Projected employment in 2034: 40,600**
- **Projected growth, 2024–2034: 42% (12,000 jobs)**
- **Projected openings: about 4,100 per year on average**

These are national occupational statistics, not guaranteed starting pay. They do not represent electrician wages and should not be used to price electrical work outside the installer classification. Local pay varies with experience, location, union status, overtime, travel, project type, credentials, and employer.

United States — non-government market supplement

Indeed's U.S. Solar Installer salary page, updated July 17, 2026, displayed an average base salary of **\$26.12 per hour**, with a displayed low of **\$16.37** and high of **\$41.68 per hour**, based on roughly 2,900 salaries from job postings over the prior 36 months. It separately displayed an overtime indicator of **\$7,500 per year**.

This is a non-government, title-specific market estimate. It is not an official occupational statistic, a guaranteed starting wage, or a promise of overtime. Check current local postings and the methodology shown on the live page.

Canada

Government of Canada Job Bank reports national wages for the NOC 73200 solar-panel-installer mapping, updated November 19, 2025:

- **CAD \$18.65 per hour low**
- **CAD \$26.00 per hour median**
- **CAD \$40.00 per hour high**

Use Job Bank's regional filters. Do not blend these installer-group wages with Construction Electrician wages or assume the same scope of work.

A practical 12-week starter plan

Weeks 1–2: Choose the role you want. Compare installer, racking, electrical-helper, construction, and maintenance postings. Mark which duties are mechanical and which require electrical qualifications.

Weeks 3–4: Learn basic PV terminology, construction measurement, drawing symbols, module and racking vocabulary, hazard communication, and documentation. Use approved introductory material; do not practice on energized or illuminated PV circuits.

Weeks 5–6: Compare employer training, public technical programs, apprenticeships, workforce funding, SENA or the relevant national vocational institution, and financial-aid eligibility. Verify costs and recognition.

Weeks 7–8: Complete required employer-approved safety orientation for the setting. This may address fall prevention, ladders or lifts, PPE, heat, electrical awareness, material handling, and stop-work procedures. A course does not authorize tasks outside your role.

Weeks 9–10: Under supervision, build low-risk experience in staging, measurement, inventory, documentation, approved tool use, and mechanical assembly within authorization.

Weeks 11–12: Apply for helper, trainee, apprentice, and entry installer roles. Prepare examples showing careful measurement, safe teamwork, reliable attendance, accurate records, and willingness to stop when a condition is unclear.

Building experience without already having the title

Transferable experience can come from construction labor, roofing support, electrical-helper work, carpentry, warehouse and materials handling, telecom installation, facilities maintenance, field service, manufacturing, landscaping, or supervised vocational projects.

Describe the work precisely. Useful evidence includes accurate layout, safe lifting, following drawings, low rework, tool accountability, jobsite housekeeping, documentation, customer communication, and completion of recognized training. Do not claim responsibility for design, electrical connection, inspection, commissioning, or code compliance if you only assisted or observed.

Advancement

Progression may include experienced installer, crew lead, site supervisor, operations-and-maintenance technician, quality or commissioning assistant, estimator, project coordinator, technical sales support, electrician apprenticeship, construction electrician, inspector, designer, or engineering-related work. Each step has its own experience, education, credential, and legal requirements.

Ask employers what advancement actually requires. A short solar course may help with entry but does not replace an electrical apprenticeship, engineering education, inspection authority, or a jurisdictional license where one is required.

Using AI responsibly

AI tools may help explain general terminology, organize study notes, compare job postings, draft interview questions, or build a non-sensitive learning checklist. They must not replace:

- site or structural assessment;
- electrical or PV system design;
- code and regulation interpretation;
- conductor, protection, grounding, bonding, or equipment sizing;
- fall-protection or roof-access planning;
- lockout/tagout or electrical-isolation procedures;
- commissioning, inspection acceptance, or utility interconnection procedures;
- manufacturer instructions; or
- decisions by the licensed, qualified, registered, or authorized people responsible for the work.

Verify AI-generated technical claims and citations against current approved sources. Never act on an AI answer when the site condition, equipment rating, electrical state, or governing rule is uncertain.

Cybersecurity and privacy basics

Solar projects can involve customer addresses, roof and electrical photographs, site drawings, utility account data, equipment serial numbers, inverter and monitoring portals, Wi-Fi credentials, production data, schedules, pricing, permits, and identity documents.

Follow employer rules for approved devices, storage, access, passwords, multifactor authentication, photographs, customer consent, file sharing, and public AI tools. Do not put customer data, credentials, proprietary drawings, security details, or utility documents into a public AI service without authorization.

Verify unusual requests to change bank details, reset a monitoring account, share credentials, transfer confidential drawings, or buy equipment through an approved second channel. Report lost devices, suspicious links, unexpected login prompts, and unauthorized access promptly.

Avoid scams and weak training offers

Be cautious when a provider, recruiter, or employer:

- guarantees a job, license, apprenticeship, permit, interconnection approval, or specific wage;
- says a short course makes every graduate a licensed electrician or independent solar contractor;
- will not identify the recognized credential, regulator, employer partners, total cost, or refund terms;
- pressures you to borrow or pay immediately;
- encourages unsupervised roof, fall-protection, service-panel, connector, energized-circuit, commissioning, or meter practice;
- asks you to pay for a job offer, equipment shipment, or background check with gift cards, cryptocurrency, or an unusual transfer; or
- requests identity or banking information before you can independently verify the organization.

Check claims through the official employer, apprenticeship authority, regulator, public training institution, workforce agency, or utility.

When to pause before spending money

Pause before enrolling when local employers do not repeatedly request the credential, public or employer-paid options may cover the same skill, financial-aid eligibility is unclear, the provider cannot explain the difference between mechanical installation and regulated electrical work, hands-on work lacks proper supervision, or total costs and refund terms are not in writing.

Compare likely entry wages with tuition, tools, transport, unpaid time, exam fees, and any loan repayment. Training may improve preparation, but it does not guarantee admission, employment, wages, certification, licensing, inspection approval, utility interconnection, or earnings.

Current sources

Official/public sources:

- U.S. Bureau of Labor Statistics — Solar Photovoltaic Installers: <https://www.bls.gov/ooh/construction-and-extraction/solar-photovoltaic-installers.htm>
- O*NET OnLine — Solar Photovoltaic Installers (47-2231.00): <https://www.onetonline.org/link/summary/47-2231.00>
- OSHA — Solar Energy: <https://www.osha.gov/green-jobs/solar/>
- OSHA — Solar Energy, Electrical: <https://www.osha.gov/green-jobs/solar/electrical>
- OSHA — Solar Energy, Falls: <https://www.osha.gov/green-jobs/solar/falls>
- OSHA — 29 CFR 1926.501: <https://www.osha.gov/laws-regs/regulations/standardnumber/1926/1926.501>

- Apprenticeship.gov: <https://www.apprenticeship.gov/>
- Federal Student Aid — FAFSA student steps: <https://studentaid.gov/articles/fafsa-student-steps/>
- U.S. Department of Labor — WIOA guidance TEGL 07-25: <https://www.dol.gov/agencies/eta/advisories/tegl-07-25>
- Government of Canada Job Bank — Solar panel installer summary: <https://www.jobbank.gc.ca/marketreport/summary-occupation/27267/ca>
- Government of Canada Job Bank — Solar panel installer wages: <https://www.jobbank.gc.ca/marketreport/wages-occupation/27267/ca>
- Red Seal Program — Construction Electrician: <https://red-seal.ca/eng/trades/const-elect.shtml>
- SENA/Betowa — Mantenimiento e instalación de sistemas solares fotovoltaicos: <https://betowa.sena.edu.co/oferta/mantenimiento-e-instalacion-de-sistemas-solares-fotovoltaicos?modality=P&programId=72121>
- SENA/Betowa — Diseño y montaje de sistemas solares fotovoltaicos: <https://betowa.sena.edu.co/oferta/disenio-y-montaje-de-sistemas-solares-fotovoltaicos?offertype=company&programId=178494>
- Colombia Ministry of Mines and Energy — RETIE: <https://www.minenergia.gov.co/es/misional/energia-electrica-2/reglamentos-tecnicos/reglamento-t%C3%A9cnico-de-instalaciones-el%C3%A9ctricas-retie/>
- OIT/Cinterfor — institutional network: <https://www.oitcinterfor.org/red-institucional>
- OIT/Cinterfor — vocational-training network: <https://www.oitcinterfor.org/recursos/redIFP>

Non-government market supplement:

- Indeed — Solar Installer salary in the United States: <https://www.indeed.com/career/solar-installer/salaries>

Important limitations

This guide provides general educational and career-navigation information. It is not legal, electrical, engineering, structural, occupational-health, financial, or licensing advice. It is not a site-specific safety plan, code determination, electrical design, fall-protection plan, commissioning procedure, inspection decision, or utility-interconnection approval.

Training, wages, classifications, codes, regulations, trade rules, permits, funding, programs, and openings can change. Verify the current local source before acting.

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